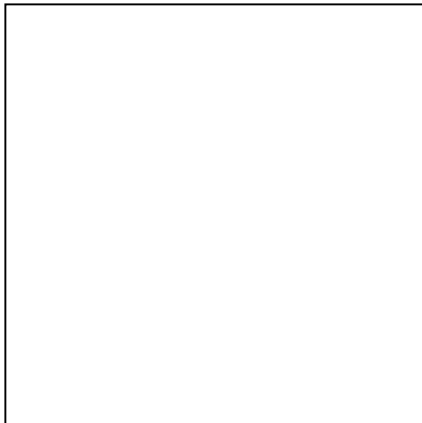


Student number:



EEE3094S 2023: Class Test 2

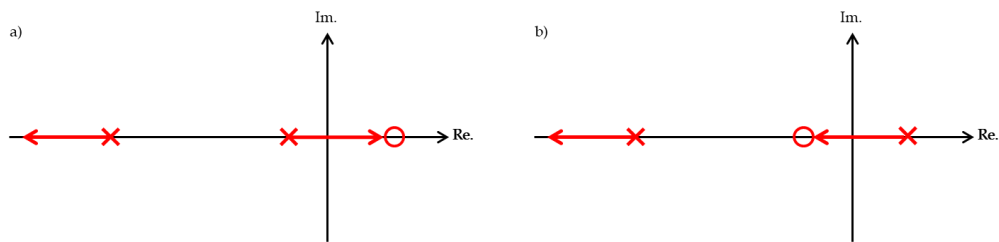
- The test is open book
- Answer on the test sheet. If you run out of space, you may attach additional pages, but please note that you have done so (e.g. write “see attached pages” by the question) and make sure they are marked with your student number.
- There are four questions totalling 50 marks.
- You have 90 minutes to complete the test.
- The last page is an information sheet.
- The information sheet includes a photo of my cat, Mouse. You are invited to draw a picture of her in the space below if you finish/ give up early and are bored. Doing so will not officially get you extra marks, but it will probably make me and the tutor marking your test happy, which might make a difference in your favour. **Also, the “best” Mouse drawing will gain the artist an unspecified reward.**



Class Test 1 - Memo

Question 1 [6]

The Nyquist plot of an unknown system encircles -1 once in a clockwise direction. What does this tell you about the stability of the closed-loop system, if the root locus of the system is as shown in each of the following cases? [2 x 3 marks]



Solution

- a) The system has no open-loop poles in the right-half plane, so substituting $N = 1$ and $P = 0$ into the stability criterion $N = Z - P$ gives $Z = 1$ closed-loop pole in the right-half plane. The closed-loop system is therefore unstable.
- b) The system has one open-loop pole in the right-half plane, so $P = 1$ and therefore $Z = 0$. There are no closed-loop poles in the right-half plane, so the closed-loop system is stable.

Question 2 [22]

The transfer function relating the position to the field voltage of a field-controlled DC motor is

$$G(s) = \frac{A}{s(1 + s\tau_m)(1 + s\tau_e)}$$

where A is the gain of the motor, τ_m is a mechanical time constant related to the mechanical damping and inertia of the motor shaft, and τ_e is the electrical time constant related to the resistance and inductance of the field coil.

Because the mechanical time constant is often much larger than the electrical one, the reduced-order model $G_r(s) = \frac{A}{s(1 + s\tau_m)}$ is often a reasonable simplification.

You may use $A = 1$, $\tau_m = 0.5$ and $\tau_e = 0.05$.

- a) Does the reduced-order model behave the same way as the original model with respect to closed-loop stability? You may support your answer with rough sketches of a suitable plot. [4]

You decide to place this motor in feedback with a proportional controller to have it track a position input.

- b) Evaluate whether it is possible to meet the following design specifications. Give brief explanations supporting your decisions:
- i) Tracking of a position with >90% accuracy. [2]
 - ii) Perfect correction of a constant offset in the output position. [2]
 - iii) Settling time less than the mechanical time constant. [2]
- c) A design specification states that the response of the system should have no more than 10% overshoot. Using the reduced-order model, identify a position for the closed-loop poles that gives the fastest possible settling time and 10% overshoot. Find the value of K needed to place the closed-loop poles there. Would you recommend using this gain value? [8]
- d) You wish to upgrade the controller so the motor can track a constant velocity without error. Would this be feasible to achieve with an integral controller of the form $C(s) = \frac{K}{s}$? (You can use either model in your analysis. You may also wish to support your analysis with a suitable sketch.) [4]

Solutions:

- a) No: the reduced-order model will not become unstable at large gains, while the original model will.

This can be illustrated by a root locus, where the reduced-order model will have vertical asymptotes, and the original model will have asymptotes at $\pm \frac{\pi}{3}$ that take the poles into the right-half plane.

Alternatively, a rough Bode plot of each model will show that the reduced-order model has infinite gain margin (the phase asymptotically approaches $-\pi$), while the original model crosses $-\pi$ and therefore has a finite gain margin.

- b) i) Yes. This is a type 1 system, so it will track a constant position with 100% accuracy even with proportional control.
ii) Yes. The offset can be represented by a constant output disturbance, which would also be rejected perfectly by a type 1 system.

iii) No. Because of the pole at the origin, the dominant closed-loop poles will always be to the right of (slower than) the pole with this time constant.

- c) The root locus of $G_r(s)$ shows that the poles will have a real part of $\sigma = -1$ when the system is critically damped, so this is the fastest possible position for the closed-loop poles. [1]

Commented [HW1]: Would this not be -1 (-1/0.5 divided by 2)?

The percentage overshoot is given by $\%OS = 100e^{\left(\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}\right)}$, so for 10%, we can solve to find the required damping ratio:

$$\ln(0.1) = \frac{-\zeta\pi}{\sqrt{1-\zeta^2}}$$

$$\therefore \ln(0.1) \sqrt{1-\zeta^2} = -\zeta\pi$$

$$\therefore \ln(0.1)^2 (1-\zeta^2) = \zeta^2 \pi^2$$

$$\therefore \ln(0.1)^2 = (\ln(0.1)^2 + \pi^2) \zeta^2$$

$$\therefore \zeta = \sqrt{\frac{\ln(0.1)^2}{\ln(0.1)^2 + \pi^2}} = 0.5912 \approx 0.6 \text{ [1]}$$

The damping ratio is related to the angle of the poles with respect to the real axis by $\cos(\theta) = \zeta$, so $\frac{|\sigma|}{\omega_n} = 0.6$, meaning the magnitude (ω_n) of the pole is $\frac{1}{0.6} = 1.67$. Using Pythagoras, the imaginary part of the pole is then $\sqrt{\omega_n^2 - \sigma^2} = 1.33$. [2]

The value of the open-loop function at $s = -1 + j1.33$ is -0.72.

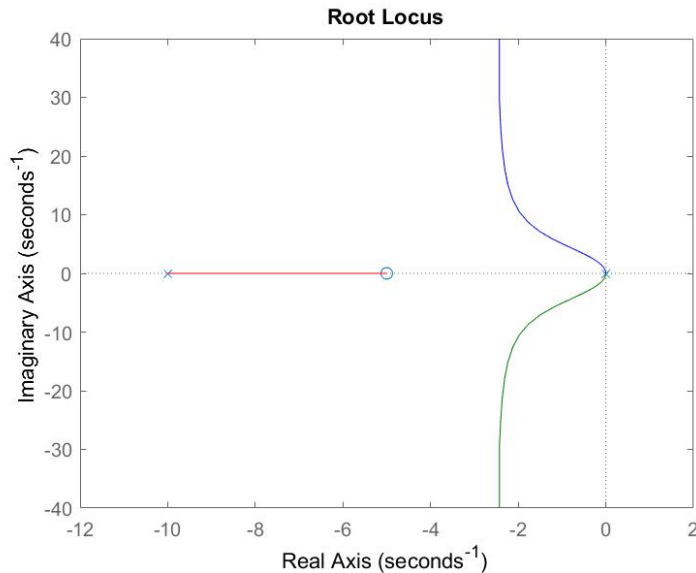
The necessary gain is therefore $K = \frac{1}{0.72} = 1.3889$. [2]

It is not recommended to use this gain if the maximum allowed overshoot is 10% however, as uncertainties in the model or other fluctuations could cause the overshoot to exceed this value if the poles are placed precisely on the boundary of the specification. This should rather be seen as an upper limit. [2]

- d) The integral controller will increase the type number of the system from 1 to 2, which is sufficient to track a ramp without error, but attempting to implement this controller will result in an unstable closed-loop system, as the direction of the asymptotes show that the two poles at the origin will move into the right-half plane.

Question 3 [10]

Consider the root locus of a system shown below:



Based on this plot, decide whether the given statements about the system are true or false. Support your answers with brief explanations.

- a) The system has infinite gain margin when placed in feedback. [2]
- b) The system will not be affected by input or output disturbances that can be modelled as a constant velocity offset. [3]
- c) The system will behave more like a second-order system with increasing gain. [2]
- d) Adding another zero between -4 and the origin will cause the system to behave like an overdamped second-order system at high gains. [3]

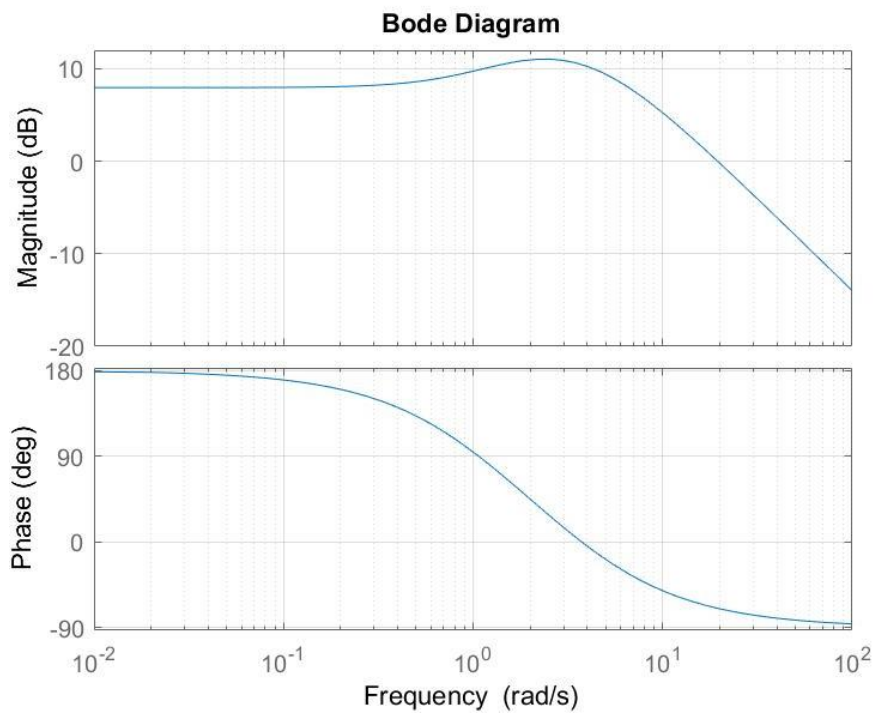
Solutions:

- a) TRUE. The poles do not cross into the right-half plane at any gain value.
- b) FALSE. While this is true for output disturbances due to the system being type 2, the two integrators would have to be included in the controller transfer function rather than the plant to affect input disturbances.
- c) FALSE. The non-dominant pole moves closer to the dominant pole pair as gain increases.
- d) TRUE. Adding another zero in this region of the real axis will cause the poles at the origin to break into the real axis between the zeros. The non-dominant pole will move out to infinity as the gain increases, so two real poles will dominate the response.

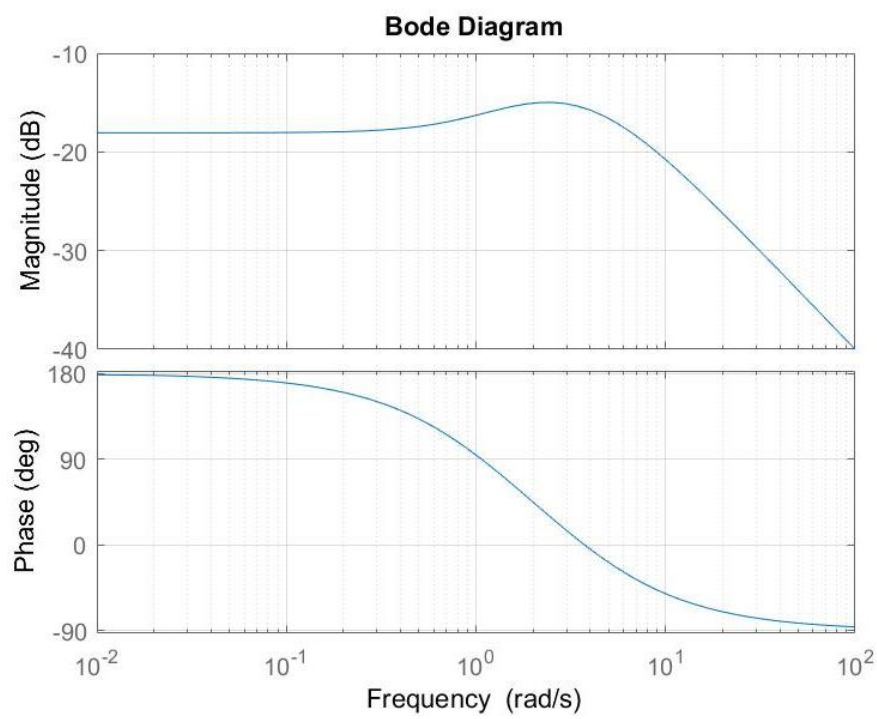
Question 4 [12]

The given Bode plots show the frequency responses of systems that are stable in open loop. For each of these plots, determine the gain and phase margin of the system, and annotate the plot to illustrate your working (i.e. indicate the gain and phase crossovers where applicable). Evaluate whether the closed-loop system is stable, and if so, whether a proportional controller could cause it to become unstable. [3 x 4 marks]

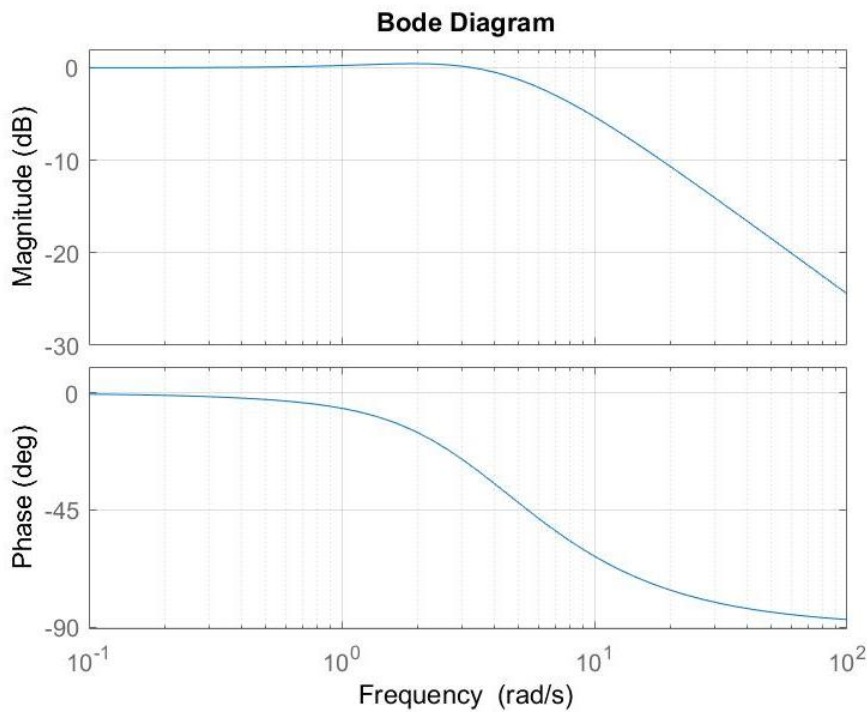
a)



b)



c)



Solutions:

- a) This system has zero gain or phase margin, as the gain is already greater than 1 at the phase crossover frequency. The closed-loop system is therefore unstable.
- b) The gain of the system is around -18 dB at the phase crossover, so there is roughly 18 dB of gain margin. The gain doesn't cross 0 dB, so the phase margin is infinite. The system is stable in closed loop, but could become unstable with a large enough proportional gain.
- c) The system has infinite gain margin as the phase does not cross any odd multiples of 180 degrees. The phase margin is 180 degrees, as the phase is zero at the gain crossover. The system is therefore stable in closed loop, and cannot be destabilized by a proportional gain.

Information

$$\frac{A\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\omega_d = \omega_n \sqrt{1 - \zeta^2}$$

$$\cos \theta = \zeta$$

$$t_p = \frac{\pi}{\omega_d}$$

$$y_p = A \left(1 + e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}} \right)$$

$$t_r = \frac{\pi - \theta}{\omega_d}$$

$$t_{k\%} = \frac{\ln \left(\frac{k}{100} \sqrt{1 - \zeta^2} \right)}{-\zeta\omega_n}$$

$$\%OS = 100e^{\left(\frac{-\zeta\pi}{\sqrt{1-\zeta^2}} \right)}$$

Laplace transforms of signals:

$y(t)$	$Y(s)$	Comments
$\sigma(t)$	$\frac{1}{s}$	unit step
$t\sigma(t)$	$\frac{1}{s^2}$	unit ramp
$e^{-at}\sigma(t)$	$\frac{1}{(s+a)}$	decaying exponential if $a > 0$
$t^n e^{-at}\sigma(t)$	$\frac{n!}{(s+a)^{(n+1)}}$	n^{th} order exponential
$\sin(\omega t)\sigma(t)$	$\frac{\omega}{s^2 + \omega^2}$	sinusoid
$\cos(\omega t)\sigma(t)$	$\frac{s}{s^2 + \omega^2}$	<u>cosinusoid</u>
$e^{-at}\sin(\omega t)\sigma(t)$	$\frac{\omega}{(s+a)^2 + \omega^2}$	damped sinusoid
$e^{-at}\cos(\omega t)\sigma(t)$	$\frac{s+a}{(s+a)^2 + \omega^2}$	damped <u>cosinusoid</u>



Fig. 1 Mouse on a roll